

# Austin Lutterbach

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## EDUCATION

### Cornell University

- Degree: Masters of Engineering
- Concentration: Applied Operations Research

GPA 3.80  
Aug 2020 - May 2021

### Purdue University

- Degree: BS in Math and Statistics

GPA 3.73  
Aug 2017 - May 2020

## SKILLS

**Frameworks:** Python (8+ yrs), LaTeX (8+ yrs), Markdown (8+ yrs), LlamaIndex (2+ yrs)

**Software/COTS:** Docker (3+ yrs), Linux (3+ yrs), Git (3+ yrs), Cursor (1 yr)

## RELEVANT EXPERIENCE

### Booz Allen Hamilton, Machine Learning Engineer/Scientist

Jan 2024 - Present

- Fine-tuned a local LLM (gemma 2b) to be used by combat medics to transcribe critical information in chaotic environments.
- Co-authored white paper detailing current trends in LLM observability functionalities.
- Developed adversarial patch attacks on object detection models to drive misclassifications.
- Developed poison patrol, a software tool that combines multiple methods of detecting poison in training data into one cohesive detection mechanism, building on latest research practices.
- Developed agentic workflows to help streamline fraud detection for USPS.
- Developing agentic workflows for OT cybersecurity applications.

### Cobec Consulting, Data Scientist

Aug 2021 - Jan 2024

- Improved failure forecast accuracy by leveraging industry-standard modeling tools (scipy, scikit-learn, tensorflow), reducing model RMSE by 35% compared to existing FAA models.
- Predicted critical parts failures in the NAS using LSTM deep learning models on time-series failure data.
- Assisted in developing Java and SQL-based database systems to parse live flight data for statistical analysis.
- Nominated for ICEAA 2023 Junior Analyst of the Year Award for analyses leading to better-informed FAA infrastructure decisions, saving millions in avoided risk mitigation costs.

### Utiliz, Machine Learning Engineer (Startup)

Jan 2022 - Sep 2023

- Led the end-to-end development of a machine learning project predicting maintenance needs for tens of thousands of water pipes in Evansville.
- Streamlined data preprocessing, model training, deployment, and maintenance using AWS Glue and SageMaker along with Docker.
- Engineered a Graph Convolutional Neural Network (GCN) with PyTorch to better capture pipe network structure and enhanced failure prediction accuracy to 79%.
- Utilized an OCR parser to extract and vectorize textual data from handwritten legal documents with Llama 2 LLM to generate automated legal reports based on historical report structures.
- Implemented a dual-layer verification mechanism by utilizing another LLM to scrutinize and validate the accuracy of the generated reports against a comprehensive database of water utility documentation, ensuring the production of high-quality and error-free outputs with optional human oversight in the final review stage.

### Laboratory for Computational Anthroinformatics, Purdue University, Research Assistant

Feb 2019 – Aug 2020

- Developed machine learning models and performed statistical analyses on archaeological data using R, exploring ancient human dietary and migratory patterns
- Conducted literature review on morphometrics and implemented methods such as generalized Procrustes into software package called “GUImorph” to help archaeologists streamline morphometric calculations on archaeological specimens.

## PROJECTS

### Summer of Open AI Research, EleutherAI

August 2026

- Supported the UniverseTBD research group in project titled “Effective Rank and Semantic Compression in Foundation Models”
- Researched methods to evaluate the “Platonic Representation Hypothesis” by trying to decouple properties of the underlying manifold with the performance of alignment evaluations.
- Discovered connection between linear probe performance restricted to embedding manifold and label curvature relative to the bending of latent space.
- Authored white paper on probe findings and submitted to ML4PS workshop as first author.